

Generation Interconnection System Impact Study Report

For Queue Project Q11-0032

Fuller Road 138 kV

March 16, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0032. The Interconnection Customer proposes a 183.3 MW wind facility interconnecting at the Fuller Road 138 kV substation in the Virginia area. The request was received on April 25, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 27 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$103,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Brookridge - Burrows Crossing Rd 138 KV is overloaded to 105.3% of its 310 MVA emergency rating for the loss of the Brindley - Carreon 138 KV. The project contributes 10.3% DFAX (18.9 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Brookridge - Burrows Crossing Rd 138 KV	Reconductor line	\$494,000	21% (\$103,000)
	Total allocated to project		\$103,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.